

IDENTIFYING INFORMATION:**NAME:** Yang, Zong-Liang**POSITION TITLE:** John A. and Katherine G. Jackson Chair in Earth System Science**PRIMARY ORGANIZATION AND LOCATION:** Department of Earth and Planetary Sciences, The University of Texas at Austin, Austin, Texas, United States**Professional Preparation:**

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Macquarie University, Sydney, NSW, Australia	PHD	08/1989 - 07/1992	Atmospheric Sciences
Melbourne University, Melbourne, Victoria, Australia	MS	04/1986 - 03/1989	Meteorology
Nanjing University of Information Science and Technology, Nanjing, Jiangsu Province, China	BS	09/1980 - 07/1984	Meteorology

Appointments and Positions

2011 - present	John A. and Katherine G. Jackson Chair in Earth System Science, Department of Earth and Planetary Sciences, The University of Texas at Austin, Austin, Texas, United States
2011 - 2021	Founding Director, Center for Integrated Earth System Science, University of Texas at Austin, Austin, TX, United States
2008 - present	Professor, University of Texas at Austin, Austin, TX, United States
2005 - 2008	Associate Professor, University of Texas at Austin, Austin, TX, United States
2001 - 2005	Assistant Professor, University of Texas at Austin, Austin, TX, United States
1997 - 2001	Research Associate Professor, University of Arizona, Tucson, AZ, United States
1995 - 1997	Senior Research Specialist, University of Arizona, Tucson, AZ, United States
1993 - 1995	Postdoctoral Research Fellow, University of Arizona, Tucson, AZ, United States

Products

1. Niu G. Y., Yang Z. L., Mitchell K. E., Chen F., Ek M. B., Barlage M., Kumar A., Manning K., Niyogi D., Rosero E., Tewari M., Xia Y. L.. The community Noah land surface model with multiparameterization options (Noah-MP): 1. Model description and evaluation with local-scale measurements. Journal of Geophysical Research-Atmospheres. 2011; 116. DOI: D12109 10.1029/2010jd015139
2. Yang Z. L., Niu G. Y., Mitchell K. E., Chen F., Ek M. B., Barlage M., Longuevergne L., Manning K., Niyogi D., Tewari M., Xia Y. L.. The community Noah land surface model with multiparameterization options (Noah-MP): 2. Evaluation over global river basins. Journal of Geophysical Research-Atmospheres. 2011; 116. DOI: D12110 10.1029/2010jd015140
3. Gent P, Danabasoglu G, Donner L, Holland M, Hunke E, Jayne S, Lawrence D, Neale R, Rasch

- P, Vertenstein M, Worley P, Yang Z, Zhang M. The Community Climate System Model Version 4. *Journal of Climate*. 2011 October; 24(19):4973-4991. Available from: <https://journals.ametsoc.org/doi/10.1175/2011JCLI4083.1> DOI: 10.1175/2011JCLI4083.1
4. Nielsen-Gammon J, Banner J, Cook B, Tremaine D, Wong C, Mace R, Gao H, Yang Z, Gonzalez M, Hoffpauir R, Gooch T, Kloesel K. Unprecedented drought challenges for Texas water resources in a changing climate: what do researchers and stakeholders need to know?. [Preprint]. 2020 June 14. DOI: 10.1002/essoar.10503372.1
 5. Souto L, Yip J, Wu W, Austgen B, Kutanoğlu E, Hasenbein J, Yang Z, King C, Santoso S. Power system resilience to floods: Modeling, impact assessment, and mid-term mitigation strategies. *International Journal of Electrical Power & Energy Systems*. 2022 February; 135:107545-. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S014206152100781X> DOI: 10.1016/j.ijepes.2021.107545
 6. Sui X, Nielsen-Gammon J, Yang Z, Niyogi D. Divergent urban storm response to convective, frontal and tropical systems. *Nature*. 2026 May 20; 653(8116):1078-1084. Available from: <https://www.nature.com/articles/s41586-026-10479-7> DOI: 10.1038/s41586-026-10479-7
 7. Li L, Yang Z, Matheny A, Zheng H, Swenson S, Lawrence D, Barlage M, Yan B, McDowell N, Leung L. Representation of Plant Hydraulics in the Noah-MP Land Surface Model: Model Development and Multiscale Evaluation. *Journal of Advances in Modeling Earth Systems*. 2021 April 16; 13(4):- . Available from: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2020MS002214> DOI: 10.1029/2020MS002214
 8. Mbarak M, Singh M, Sudharsan N, Yang Z. A deep learning bias-correction layer for land surface models: Application to soil moisture during drought and hurricane events. *Machine Learning: Earth*. 2026 May 21; :- . Available from: <https://iopscience.iop.org/article/10.1088/3049-4753/ae7115> DOI: 10.1088/3049-4753/ae7115
 9. Somnath Luitel, Manmeet Singh, Joshua Durkee, Abdullah Al Fahad, Naveen Sudharsan, Prabhjot Singh, Cenlin He, Harsh Kamath, Zong-Liang Yang, Krishnagopal Halder, Sandeep Juneja, Parthasarathi Mukhopadhyay, Saptarishi Dhanuka, Amit Kumar Srivastava. AirCast-SR: A Foundation Model for Kilometer-Scale Atmospheric Super-Resolution via Latent Consistency Diffusion. 2026. Available from: <https://arxiv.org/abs/2605.26130> DOI: 10.48550/ARXIV.2605.26130
 10. Singh M, Acharya N, Jamshidi S, Jiao J, Yang Z, Coudert M, Baumer Z, Niyogi D. DownScaleBench for developing and applying a deep learning based urban climate downscaling- first results for high-resolution urban precipitation climatology over Austin, Texas. *Computational Urban Science*. 2023 May 31; 3(1):- . Available from: <https://link.springer.com/10.1007/s43762-023-00096-9> DOI: 10.1007/s43762-023-00096-9

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

Certified by Yang, Zong-Liang in SciENcv on 2026-05-31 20:55:30